

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. **Name of the Academic Unit: Department of Electronics and Electrical Communication Engineering**

2. **Subject Name: Communication-II L-T-P: 3-1-0 Credits: 4**

3. **Pre-requisites: - Communication-I**

4. **Syllabus and reference books:**

Syllabus: Probability and Random Process, Source and Channel waveforms, Introduction to Information theory, Source Coding, Signal Detection, Channel Capacity, Channel coding, Wireless channel, Multiuser communication, Introduction to Communication Networks.
Reference Books: 1. Principles of Digital Communication: Robert Gallager, Cambridge University Press, 2008. 2. Digital Communications: John Proakis, McGraw Hill, 2001. 3. Principles of Communication Engineering: Wozencraft and Jacobs, Waveland Press, 1990. 4. Wireless Communications: Andrea Goldsmith, Cambridge University Press, 2005

5. **Lecture-wise break-up:**

Sl. No.	Topic	No. of lectures
1.	Probability and Random Process: Probability space, random variable, distribution, moments, inequalities and limit theorems, conditional probability, random processes, moment functions, stationarity, ergodicity, Wiener-Khinchine theorem, filtering of random processes	6
2.	Source and Channel waveforms: Signal spaces, basis, Gram-Schmidt orthogonalization, time limitation and band limitation, Eigen analysis, dimensionality theorem	4
3.	Introduction to Information theory: Entropy, Joint Entropy, Conditional Entropy, Relative Entropy and Mutual Information, Chain rule, entropy rate, Data-Processing Inequality, Asymptotic Equipartition Property (AEP)	4

4.	Source Coding: Fixed and Variable Length Codes, Kraft Inequality, Huffman Codes, Shannon-Fano-Elias Coding, Lempel-Ziv Algorithm	5
5.	Signal Detection: AWGN channel, K-L expansion, MAP/ML detection, optimal correlation receiver, probability of error, union bound and lower bound of probability of error, Sequence detection, trellis, Viterbi algorithm, trade-offs	6
6.	Channel Capacity: Channel abstraction, Symmetric Channels, Capacity, Jointly Typical Sequences, Channel Coding Theorem	5
7.	Channel coding: Hamming code, block code, cyclic code, convolutional code	5
8.	Wireless channel: Channel modelling, pathloss, shadowing, channel impulse response, fading, delay profile, doppler power spectrum, stochastic channel models: Rayleigh, Rician and Nakagami channel models, Markov models for fading.	5
9.	Multiuser communication: Multiple access techniques: time division, frequency division, and spatial division multiple access schemes, principles of orthogonal frequency division multiple access (OFDMA) and code-division multiple access (CDMA), spread spectrum communication: direct sequence spread spectrum (DSSS), frequency hopping spread spectrum (FHSS), receiver design for multiuser communication.	5
10.	Introduction to Communication Networks: Network architecture and layering, OSI layer, statistical multiplexing, basics of switching and routing. Multiple access and resource allocation in networks, ALOHA, carrier sensing, CSMA, CSMA/CD, CSMA/CA	3
Total number of hours		48